SAVEETHA SCHOOL OF ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Department of Computer Science and Engineering

CSA47 – Deep Learning

ASSIGNMENT 1 – CO1

Mathematical Foundations and Machine Learning Principles for Healthcare Risk Prediction

CO1: Analyse the foundational mathematical concepts, including linear algebra and probability, and core machine learning principles such as bias–variance trade-offs and supervised/unsupervised learning algorithms.

Student Name	THAMIZHARASAN S	**Register Number**	192424087
Department	Computer Science and Engineering	**Academic Year**	2026–2027
Faculty Name	JAGATHEESAN A	**Nature of Assignment**	Individual
Duration	1 Week	**Total Marks**	50
Date of Submission	02/11/2026	**Date of Evaluation**	03/11/2026

Mark Distribution

Task	Task Title	Max	Marks Awarded
1	Feature Representation and Normalization – Linear Algebra	5	
2	Vector Representation and Matrix Operations	2	
3	Probability and Conditional Probability	6	
4	Bayes' Theorem and Bayesian Classification	4	
5	Entropy and Information Gain	3	
6	Numerical Computation and Distance Measurement	3	
7	Bias–Variance Trade-off	4	
8	Overfitting, Underfitting and Model Capacity	4	
9	Supervised Learning – Model Evaluation	3	
10	Unsupervised Learning – Clustering Analysis	4	
11	Comparison of Learning Approaches	4	
	Sub Total	42	
12	Design Challenge – Selecting and Adapting a Learning Strategy	8	
	TOTAL	**50**	

Student Signature	**Faculty Signature**	**Program Director**

DEEP LEARNING (CSA47)
ASSIGNMENT 1 – CO1

CO1: Mathematical foundations (linear algebra, probability and information theory, numerical computation, Bayesian statistics) and core machine learning principles (capacity, overfitting/underfitting, bias–variance trade-off, supervised and unsupervised learning).

Nature of Assignment: Individual

Duration: 1 Week

Total Marks: 50

Submission Date: 02/11/2026

Assignment Title: Healthcare Risk Prediction (Total Marks – 50)

A hospital plans to develop an intelligent system to predict **disease risk** using patient **age, blood pressure, glucose level, cholesterol and family history**. The dataset contains **labelled** patient records as well as **unlabelled** records.

You are required to analyse the data, select suitable learning approaches and **justify decisions using mathematical reasoning**.

Design Challenge (used in Task 12):

The hospital initially analyses 10,000 patient records. It must now process 1,000,000 records, predict disease risk, discover patient groups, and handle missing values and differences in feature scales.

Decide whether to retain, modify or redesign the analytical strategy.

How this report is organised

Each task gives the question as set, followed by my answer under **Ans:**. Every calculation in Tasks 1–10 and 12 is worked out step by step in the text and checked by one Python program, healthcare_ml.py (Appendix A), which also draws all figures; python healthcare_ml.py reproduces every number. Tables and figures are numbered in order. Units are: age in years, blood pressure = systolic pressure in mmHg, glucose and cholesterol in mg/dL.

Data. The three-patient and six-patient tables are small, made-up examples chosen so that the arithmetic can be checked by hand; they are not real patient records. The numbers given in the questions themselves (1,000 patients, 10% prevalence, 92/100 correct and so on) are used exactly as set. Where I had to assume a number the question does not give (for example the glucose split in Task 5), I say so where it is used.

Relation to CO1. The report follows one thread. Tasks 1, 2 and 6 show how **scale and geometry** (vectors, matrices, distances) decide what a model "sees". Tasks 3–5 show how **probability and information theory** turn uncertain evidence into decisions. Tasks 7 and 8 explain **generalisation** (bias, variance, capacity). Tasks 9–11 compare the **learning paradigms**, and Task 12 combines all of them in one design for a hospital that is 100 times larger than the one first studied.

TASK 1 – Patient Data Representation and Normalization 5 Marks

A hospital records age, blood pressure, glucose level and cholesterol for three patients. These features have different numerical scales.

a) Explain the likely effect of using the raw features without centering or normalization.

b) Represent the patient data as a feature matrix and apply a suitable normalization procedure using a small numerical example.

c) Explain why normalization is important for distance-based and gradient-based learning methods.

Evidence Expected

Feature matrix, normalization procedure, numerical verification, explanation of scale differences and interpretation.

Ans:

1. The data as a feature matrix

Each patient is one **row** and each feature one **column**, so the three patients form a matrix $X \in \mathbb{R}^{3\times4}$ with entries x_{ij} (patient i, feature j). The columns are age (years), blood pressure (mmHg), glucose (mg/dL) and cholesterol (mg/dL).

Table 1: Feature matrix X (rows = patients, columns = features)

Patient	Age (years)	Blood pressure (mmHg)	Glucose (mg/dL)	Cholesterol (mg/dL)
P1	30	110	90	170
P2	45	130	110	200
P3	60	150	170	260

$$X = [\ 30\ \ 110\ \ 90\ \ 170\ ;\ \ 45\ \ 130\ \ 110\ \ 200\ ;\ \ 60\ \ 150\ \ 170\ \ 260\ ]\ \ \text{(rows = P1, P2, P3)}$$

2. (a) Effect of using the raw features

The four columns are measured in different units and their numerical sizes differ (cholesterol values are about 4 to 7 times larger than age values, and the glucose and cholesterol columns vary more than the age and blood-pressure columns). Using them as they are has three effects:

1. **The large-scale features dominate any distance.** Table 6 splits the squared Euclidean distance between P2 and P3. Glucose and cholesterol give 3,600 each, together **92.0%** of the total 7,825, while age gives only 2.9% and blood pressure 5.1%. A model that compares patients by distance would in effect compare only glucose and cholesterol, although a 15-year age gap is clinically important.
2. **Not centred, so the features share a large common component.** All raw values are positive and far from zero (means 45, 130, 123.3 and 210). Weights and the intercept of a learning model then compete to explain the same offset, so the loss surface is stretched and gradient descent zig-zags (see part c).
3. **Weights cannot be compared and arithmetic can go wrong.** A weight of 0.01 on cholesterol and 0.01 on age mean different things because the features have different units. Large raw inputs can also push exp(·) in a sigmoid or softmax into overflow or saturation.

3. (b) Normalization procedure and numerical example

I use **standardization (z-score)**, which both **centres** each column (subtracts its mean) and **scales** it (divides by its standard deviation):

$$z_{ij} = (x_{ij} - \mu_j) / \sigma_j, \quad \mu_j = (1/n) \Sigma_i x_{ij}, \quad \sigma_j = \sqrt{((1/n) \Sigma_i (x_{ij} - \mu_j)^2)}$$

Here n = 3 and σ_j is the population standard deviation (the same convention as scikit-learn's StandardScaler). In a real study μ_j and σ_j are computed from the **training** patients only and then reused for validation, test and new patients.

Table 2: Column means and standard deviations

	Age	Blood pressure	Glucose	Cholesterol
Mean μ_j	45.00	130.00	123.33	210.00
Std. deviation σ_j	12.25	16.33	33.99	37.42

Worked entries. Age of P1: (30 − 45)/12.247 = **−1.2247**. Glucose of P3: (170 − 123.33)/33.99 = 46.67/33.99 = **+1.3728**. Cholesterol of P2: (200 − 210)/37.42 = **−0.2673**.

Table 3: Standardized matrix Z (z-scores)

Patient	Age	Blood pressure	Glucose	Cholesterol
P1	−1.2247	−1.2247	−0.9806	−1.0690
P2	+0.0000	+0.0000	−0.3922	−0.2673
P3	+1.2247	+1.2247	+1.3728	+1.3363

Numerical verification. After the transformation every column must have mean 0 and standard deviation 1. For age: (−1.2247 + 0 + 1.2247)/3 = 0 and ((−1.2247)^2 + 0 + 1.2247^2)/3 = (1.5 + 0 + 1.5)/3 = 1. The program confirms the same for all four columns:

Table 4: Verification of the standardized columns

Check	Age	Blood pressure	Glucose	Cholesterol
Column mean of Z	0.0000	0.0000	0.0000	0.0000
Column std. deviation of Z	1.0000	1.0000	1.0000	1.0000

Second procedure: min–max scaling. Another common choice maps every column to [0, 1]: x′ = (x − min)/(max − min). It is used in Tasks 2 and 6 because probabilities and weighted scores are easy to read on a 0–1 scale.

Table 5: Min–max normalized matrix

Patient	Age	Blood pressure	Glucose	Cholesterol
P1	0.000	0.000	0.000	0.000
P2	0.500	0.500	0.250	0.333
P3	1.000	1.000	1.000	1.000

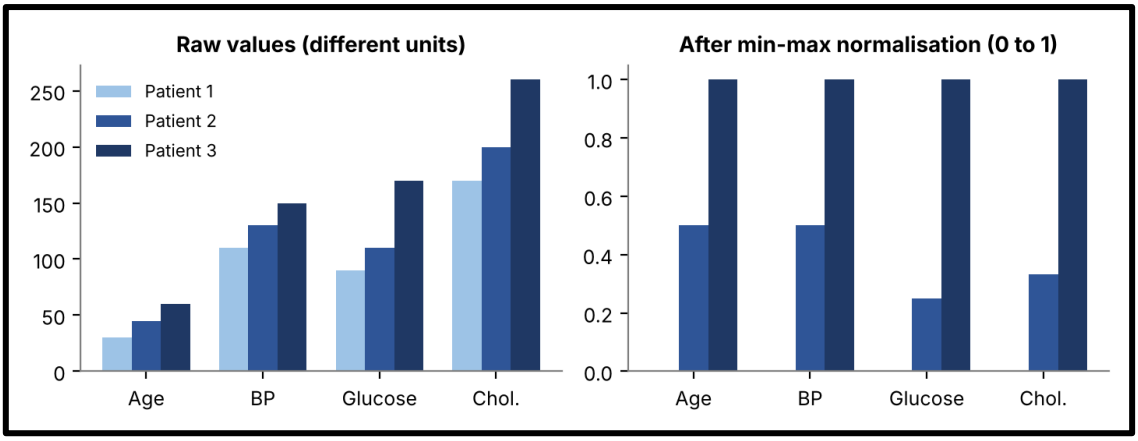


Figure 1: The same three patients before and after min–max normalization

Interpretation. In Z, P1 is below the cohort average on every feature, P2 is close to average (age and blood pressure exactly 0) and P3 is above average on every feature, most clearly in glucose (+1.37 standard deviations). The values now say "how unusual is this patient for this feature", which is comparable across features. The effect on distance is shown below: after standardization the P2–P3 distance falls from 88.46 to 2.95 units, and the four features contribute between 17% and 36% each instead of 3% to 46%.

Table 6: Share of the squared P2–P3 distance contributed by each feature

Feature	Raw squared difference	Raw share	Standardized share
Age	225	2.9%	17.3%
Blood pressure	400	5.1%	17.3%
Glucose	3,600	46.0%	35.9%
Cholesterol	3,600	46.0%	29.6%
Total / distance	**7,825 → d = 88.46**	**100%**	**100% → d = 2.95**

4. (c) Why normalization matters

Distance-based methods (k-nearest neighbours, K-means)

These methods use $d(x, x') = \sqrt{\Sigma_j (x_j - x'_j)^2}$. Because every feature enters the sum on its raw scale, the feature with the largest numerical range decides which patients are "near". Table 6 shows this: on raw data the nearest-neighbour structure is decided by glucose and cholesterol alone. Changing the unit of a feature (glucose in mmol/L instead of mg/dL divides it by about 18) would change the neighbours without changing a single patient, which is clearly undesirable. Standardization makes the distance **unit-free** so that each feature has a comparable say.

Gradient-based methods (logistic regression, neural networks)

Training minimizes a loss $L(w)$ by repeating $w \leftarrow w - \eta \nabla L(w)$. For a linear model the gradient with respect to weight w_j is $\partial L/\partial w_j = (1/n)\ \Sigma_i(\hat{y}_i - y_i)\ x_{ij}$, so **its size is proportional to the feature's size**. A single learning rate η then cannot suit all weights: it must be small enough for the large-scale feature not to overshoot, which makes progress along the small-scale features very slow. Geometrically the loss surface becomes a long narrow valley (a large condition number of the Hessian, $\kappa = \lambda_{max}/\lambda_{min}$), and gradient descent zig-zags across it. Centring removes the common offset that couples the weights to the intercept, and scaling makes the valley round (κ close to 1) so that a single η works and convergence needs far fewer

steps. In the 1,000,000-patient simulation of Task 12, standardization lowers the condition number of the feature covariance matrix from 7.38 to 1.005. Regularization (weight decay) also penalizes all weights equally, which is fair only if the features are on the same scale.

Practical rule. Fit μ_j and σ_j (or min and max) on the training data only, store them with the model, and apply exactly the same transformation to every new patient; computing them on the test data would leak information and make the test result look better than it is.

TASK 2 – Vector Representation and Matrix Operations 2 Marks

A hospital assigns different importance weights to age, glucose level, blood pressure and cholesterol when calculating a preliminary risk score.

a) Represent one patient's feature values and the importance weights as vectors and formulate the weighted score using vector notation.

b) Use a small numerical example to calculate the score and interpret it.

Evidence Expected

Vector representation, matrix/vector formulation, numerical multiplication and interpretation.

Ans:

1. (a) Vector representation and formulation

Take the feature order (age, glucose, blood pressure, cholesterol) and use normalized values in [0, 1] (Task 1). One patient is a column vector $\mathbf{x} \in \mathbb{R}^4$; the importance weights are a vector $\mathbf{w} \in \mathbb{R}^4$ with $w_j \geq 0$ and $\Sigma\, w_j = 1$, so that the score also lies in [0, 1]. The weighted score is the **dot product**:

$$s = w^T x = \Sigma_{j=1}^{4}\, w_j\, x_j = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

For a whole hospital the n patients are stacked as rows of the feature matrix $X \in \mathbb{R}^{n\times 4}$, and **one matrix–vector product** scores everybody at once:

$$s = X\, w, \quad s \in \mathbb{R}^n \ \ (s_i = w^T x_i)$$

2. (b) Numerical example

Patient: $x = (0.50, 0.80, 0.60, 0.70)^T$. Weights (age 0.20, glucose 0.40, blood pressure 0.25, cholesterol 0.15) sum to 1.

Table 7: Weighted score of one patient

Feature	Value x_j	Weight w_j	Product $w_j\, x_j$
Age	0.50	0.20	0.100
Glucose	0.80	0.40	0.320
Blood pressure	0.60	0.25	0.150
Cholesterol	0.70	0.15	0.105
Score s		**1.00**	**0.675**

$$s = 0.20(0.50) + 0.40(0.80) + 0.25(0.60) + 0.15(0.70) = 0.675$$

Matrix form for the three patients of Task 1. Using their min–max values reordered to (age, glucose, blood pressure, cholesterol) and the same **w**:

$$s = X\, w = (\ 0,\ \ 0.375,\ \ 1\ )^T \quad \text{for rows } (0, 0, 0, 0),\ (0.5, 0.25, 0.5, 0.333),\ (1, 1, 1, 1)$$

Check for P2: 0.100 + 0.100 + 0.125 + 0.050 = 0.375. P1 (lowest on every feature) scores 0 and P3 (highest) scores 1, the extremes of the scale.

3. Interpretation

- A score of **0.675** means the patient is about two-thirds of the way from the lowest to the highest risk profile in the cohort: a moderately high preliminary risk, above the cohort midpoint of 0.5.
- The score is explainable term by term: glucose contributes 0.320 of 0.675 (**47%**) because it has the largest weight and a high value, while age contributes 15%. A clinician can see why the patient was flagged. Here the weights are expert-assigned; a learning model estimates them from data (Tasks 9–11), but the calculation is the same dot product (a single neuron computes $w^T x + b$).
- The score is meaningful only because the features were normalized first; with raw values the cholesterol term (about 200 × 0.15) would swamp the age term (about 50 × 0.20). It is a **screening aid for prioritizing review**, not a diagnosis.

TASK 3 – Probability and Conditional Probability 6 Marks

In a sample of 1,000 patients, 200 have a particular disease. Of these diseased patients, 160 have elevated glucose. Among the 800 patients without the disease, 120 have elevated glucose.

a) Calculate the probability that a randomly selected patient has elevated glucose.

b) Calculate the probability that a patient has the disease given elevated glucose.

c) Calculate the probability that a patient has elevated glucose given that the patient has the disease.

d) Explain how conditional probability can support the prioritization of patients for further testing.

Evidence Expected

Event definitions, probability calculations, conditional probability formulae, numerical solutions and healthcare interpretation.

Ans:

1. Events and the counts

Let **D** = "the patient has the disease" and **E** = "the patient has elevated glucose". The statement gives N = 1,000, |D| = 200, |D ∩ E| = 160, |¬D| = 800 and |¬D ∩ E| = 120. The remaining cells follow by subtraction (200 − 160 = 40 and 800 − 120 = 680).

Table 8: Contingency table of 1,000 patients

	Elevated glucose (E)	Normal glucose (¬E)	Total
Disease (D)	160	40	200
No disease (¬D)	120	680	800
Total	**280**	**720**	**1,000**

2. (a) Probability of elevated glucose

Two routes, which must agree. **Counting:** P(E) = 280/1,000 = **0.28**. **Law of total probability:** with P(E | D) = 160/200 = 0.80 and P(E | ¬D) = 120/800 = 0.15,

$$P(E) = P(E \mid D)P(D) + P(E \mid \neg D)P(\neg D) = 0.80(0.20) + 0.15(0.80) = 0.16 + 0.12 = 0.28$$

3. (b) Probability of disease given elevated glucose

$$P(D \mid E) = P(D \cap E) / P(E) = 0.16 / 0.28 = 160 / 280 = 0.5714 \approx 57.1\%$$

4. (c) Probability of elevated glucose given disease

$$P(E \mid D) = P(D \cap E) / P(D) = 0.16 / 0.20 = 160 / 200 = 0.80 = 80\%$$

Check with Bayes' rule: $P(D \mid E) = P(E \mid D)P(D)/P(E) = 0.80 \times 0.20 / 0.28 = 0.5714$, the same as (b). Note that P(E | D) = 80% and P(D | E) = 57.1% are different numbers: 80% of diseased patients have elevated glucose, but only 57.1% of patients with elevated glucose are diseased. Confusing the two is a common error in reading medical tests.

5. (d) Using conditional probability to prioritize patients

The probability P(D | evidence) is the quantity a hospital needs for ranking, because it tells how likely the disease is **for this kind of patient**, not for the average patient:

- **Risk is raised by the evidence.** A patient with elevated glucose has P(D | E) = 57.1% against a base rate of 20%, a risk **2.86 times higher**. A patient with normal glucose has P(D | ¬E) = 40/720 = 5.6%, which is lower than the base rate.
- **Limited testing capacity is spent where it pays most.** If only 280 tests are possible, testing the 280 elevated-glucose patients finds 160 of the 200 diseased patients (hit rate 57.1%, 80% of all cases); 280 patients chosen at random would find only about 56 (20%).
- **Ranking, not excluding.** 40 diseased patients (20%) have normal glucose, so glucose alone must not rule anyone out; update the probabilities as more evidence arrives and set the cut-off by the cost of a missed case.

TASK 4 – Bayesian Statistics for Disease Classification 4 Marks

A disease affects 10% of a population. A test is positive for 90% of patients who have the disease and for 5% of patients who do not have it.

a) Using Bayes' theorem, calculate the probability that a patient has the disease given a positive test result.

b) Explain the roles of prior probability and likelihood in the posterior probability.

c) Discuss why Bayesian reasoning is useful when medical information is uncertain.

Evidence Expected

Prior and conditional probabilities, Bayes' theorem, step-by-step calculation and interpretation.

Ans:

1. (a) Bayes' theorem applied step by step

Let D = "has the disease" and + = "test is positive". Given: prior P(D) = 0.10, so P(¬D) = 0.90; likelihoods P(+ | D) = 0.90 (sensitivity) and P(+ | ¬D) = 0.05 (false-positive rate).

$$P(D \mid +) = P(+ \mid D)\,P(D) / P(+), \quad P(+) = P(+ \mid D)P(D) + P(+ \mid \neg D)P(\neg D)$$

Table 9: Step-by-step calculation

Step	Quantity	Calculation	Value
1	True positives, P(+ | D)P(D)	0.90 × 0.10	0.090
2	False positives, P(+ | ¬D)P(¬D)	0.05 × 0.90	0.045
3	Evidence, P(+)	0.090 + 0.045	0.135
4	**Posterior, P(D | +)**	**0.090 / 0.135**	**0.6667 (66.7%)**

Natural-frequency check. Take 1,000 patients: 100 have the disease and 90 of them test positive; 900 do not, and 45 of them (5%) test positive. In total 135 patients test positive, of whom 90 are really diseased: 90/135 = 66.7%.

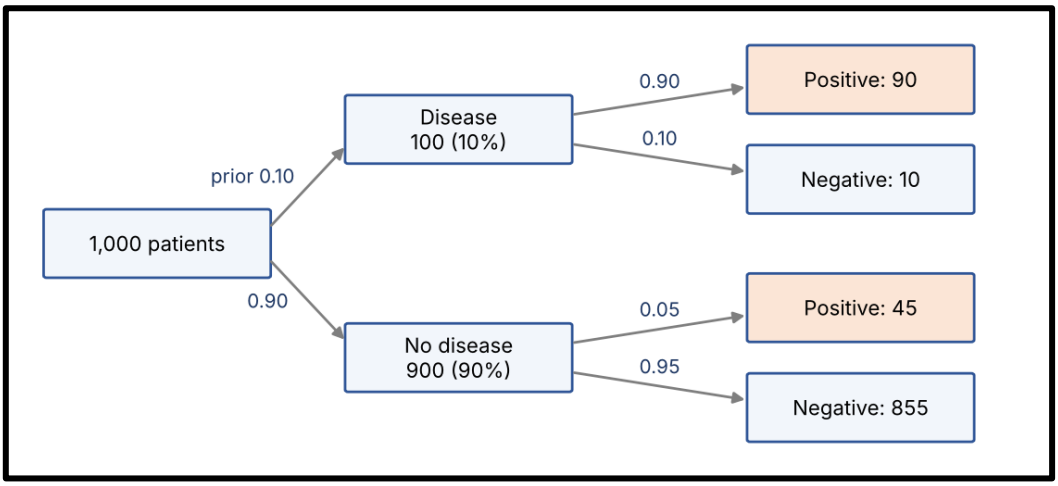


Figure 2: Probability tree for 1,000 patients; positive results are shaded

Interpretation. Although the test finds 90% of diseased patients, a positive result means only a **two-in-three** chance of disease, because the disease is rare (10%) and the large healthy group produces many false positives (45 against 90 true positives). The posterior is a different quantity from the sensitivity.

2. (b) Roles of the prior and the likelihood

Bayes' theorem says **posterior ∝ likelihood × prior**, normalized by the evidence P(+).

- **Prior P(D)** is what is believed before seeing the test: here the prevalence of 10%. It anchors the answer, so the same positive test gives different conclusions in different populations.
- **Likelihood P(+ | D)** (compared with P(+ | ¬D)) states how probable the observed result is under each hypothesis. The ratio $LR^+ = 0.90/0.05 = \mathbf{18}$ says a positive result is 18 times more likely in a diseased patient.
- **Posterior P(D | +)** combines them. In odds form: posterior odds = prior odds × LR = (0.10/0.90) × 18 = 2, i.e. a probability of 2/3, the same 66.7%.

Table 10: Same test, different prior (sensitivity 90%, false-positive rate 5%)

Prevalence (prior)	P(+)	Posterior P(D | +)	Meaning
1%	0.0585	15.4%	Screening a general population: most positives are false
10% (given)	0.135	66.7%	The case in this question
30%	0.305	88.5%	High-risk clinic: a positive is strong evidence

3. (c) Why Bayesian reasoning helps when information is uncertain

- **It turns evidence into probabilities.** Instead of a hard yes/no, the output is P(D | evidence), which can be compared with a threshold chosen from the cost of missing a case and of over-testing.
- **It updates as evidence accumulates.** The posterior becomes the new prior. If a second, independent test is also positive: P(D | ++) = 0.90(0.667) / (0.90(0.667) + 0.05(0.333)) = 0.600/0.617 = **0.973** (97.3%). The assumption of conditional independence must be checked.
- **It combines expert knowledge with data**, and copes with small samples because the prior supplies a starting point. Priors can be adjusted for each hospital's prevalence.
- **It connects to deep learning.** A Naive Bayes classifier applies this rule directly; training a network with weight decay is equivalent to maximum a-posteriori estimation under a Gaussian prior on the weights; and a network's probabilistic output should be calibrated like the posterior above.

Limits: the result is only as good as the prior and the likelihoods; wrong prevalence figures give wrong posteriors, which is why they must be estimated from representative data.

TASK 5 – Entropy and Information Gain 3 Marks

A medical dataset contains 100 patient records: 40 are classified as high-risk and 60 as low-risk.

a) Calculate the entropy of the risk classification.

b) Explain what the entropy indicates about uncertainty.

c) Explain how information gain can help select a useful feature, such as glucose level.

Evidence Expected

Entropy formula, numerical calculation, interpretation of uncertainty and explanation of information gain.

Ans:

1. (a) Entropy of the risk classification

For a classification with class probabilities p_k, the Shannon entropy (in bits) is $H = -\Sigma_k\, p_k \log_2 p_k$. Here p(high) = 40/100 = 0.4 and p(low) = 60/100 = 0.6.

$$H = -[\, 0.4 \log_2 0.4 + 0.6 \log_2 0.6 \,] = -[\, 0.4(-1.3219) + 0.6(-0.7370) \,]$$

$$H = 0.5288 + 0.4422 = 0.9710 \text{ bits}$$

2. (b) What the entropy says about uncertainty

- Entropy measures the **average uncertainty** about the class of a randomly chosen record. For two classes it ranges from **0** (all records in one class) to **1 bit** (50/50: maximum uncertainty).
- The value **0.971 bits** is very close to the maximum of 1 bit, so the 40/60 split leaves the risk class almost as uncertain as a coin toss. A useful feature must reduce this entropy.

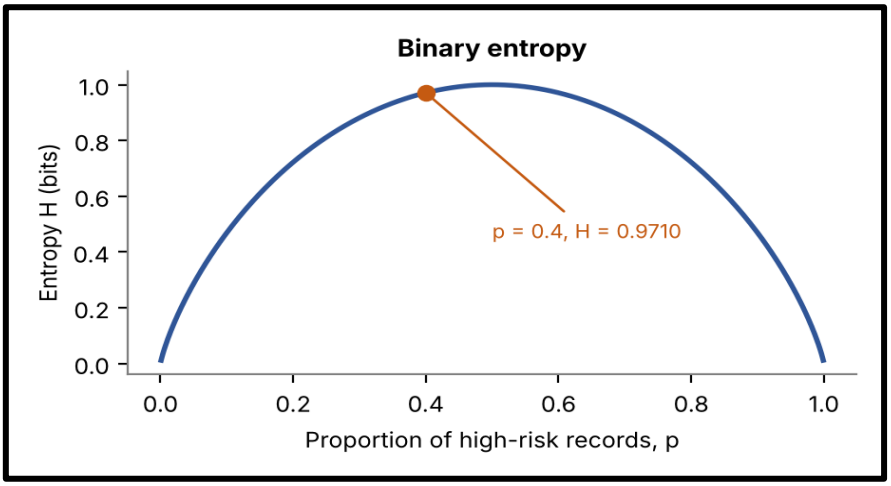


Figure 3: Binary entropy; the dataset (p = 0.4) sits near the maximum

3. (c) Information gain and feature selection

The **information gain** of a feature is the expected drop in entropy after splitting the records by that feature:

$$IG(S, \text{feature}) = H(S) - \Sigma_v \left(|S_v| / |S| \right) H(S_v)$$

Illustration with glucose (assumed split, as the question gives no table). Suppose 50 records have high glucose, of which 35 are high-risk, and 50 have normal glucose, of which 5 are high-risk (the totals 35 + 5 = 40 high-risk and 15 + 45 = 60 low-risk agree with the data set).

Table 11: Information gain of the glucose split

Subset	Records	High-risk / low-risk	Entropy $H(S_v)$
High glucose	50	35 / 15	0.8813
Normal glucose	50	5 / 45	0.4690
Weighted average after split	100		0.6751

$$IG(\text{glucose}) = 0.9710 - 0.6751 = 0.2958 \text{ bits} \quad (30.5\% \text{ of the original uncertainty removed})$$

How it is used. Compute the information gain of every candidate feature (age band, blood pressure, cholesterol, family history, glucose) and choose the one with the **largest gain** as the first split of a decision tree (ID3/C4.5) or as a top-ranked input in feature selection; a gain near 0 says nothing about risk. Caution: information gain favours features with many distinct values (gain ratio corrects this). The same idea underlies the cross-entropy loss used to train deep classifiers.

TASK 6 – Numerical Computation and Distance Measurement 3 Marks

After normalization, Patient A is represented by (0.70, 0.80, 0.60) and Patient B by (0.50, 0.75, 0.90), using glucose, blood pressure and cholesterol features.

a) Calculate the Euclidean distance between the two patients.

b) Explain how the distance can indicate similarity between patient records.

c) Explain why normalization is necessary before comparing features with different scales.

Evidence Expected

Distance formula, step-by-step calculation, similarity interpretation and explanation of normalization.

Ans:

1. (a) Euclidean distance

$$d(A, B) = \sqrt{\Sigma_j(a_j - b_j)^2} = \sqrt{[\ (a_1-b_1)^2 + (a_2-b_2)^2 + (a_3-b_3)^2\ ]}$$

Table 12: Step-by-step calculation

Feature	Patient A	Patient B	Difference a − b	Squared difference	Share of total
Glucose	0.70	0.50	+0.20	0.0400	30.2%
Blood pressure	0.80	0.75	+0.05	0.0025	1.9%
Cholesterol	0.60	0.90	−0.30	0.0900	67.9%
Sum				**0.1325**	**100%**

$$d(A, B) = \sqrt{0.1325} = 0.3640 \approx 0.364$$

2. (b) Distance as similarity

- A **small distance means similar records** (close together in feature space) and a large distance means dissimilar ones; d = 0 means identical records. Distance is therefore a measure of **dis**similarity.
- With three features in [0, 1] the largest possible distance is $\sqrt{3} = 1.732$. Here $d/\sqrt{3} = 0.210$, so A and B differ by about **21% of the maximum**: they are fairly similar.
- The squared-difference column shows **where** they differ: cholesterol accounts for 67.9% of the squared distance, glucose for 30.2%, and blood pressure is almost the same (0.05). A clinician would read this as “similar patients except that B has much higher cholesterol”.
- Distances underlie k-nearest-neighbour prediction and K-means clustering (Task 10); a value is meaningful relative to other pairs in the data set.

3. (c) Why normalization is necessary

The calculation above is meaningful only because both patients are on a common 0–1 scale. Suppose the same two patients were compared in raw units, for instance glucose 126 and 110 mg/dL, blood pressure 128 and 132 mmHg and cholesterol 200 and 260 mg/dL (illustrative values). The squared differences are $16^2 = 256$, $4^2 = 16$ and $60^2 = 3{,}600$, so $d = \sqrt{3{,}872} = 62.23$ and cholesterol alone supplies 93.0% of it. Three problems follow:

1. **Dominance by large numbers.** The feature with the largest range dominates and the others become invisible, regardless of clinical importance (the same effect as Table 6).
2. **Dependence on units.** Reporting glucose in mmol/L instead of mg/dL would change the distance and could change which patient is nearest, although nothing about the patients changed.
3. **Incomparable differences.** 20 mg/dL of glucose and 20 mmHg of blood pressure are not equally serious; scaling each feature by its own range or standard deviation makes a difference mean "a given fraction of the usual spread" everywhere.

Normalization (Task 1) must therefore be applied **before** distances are computed, using parameters learnt from the training data.

TASK 7 – Bias–Variance Trade-off 4 Marks

Three disease-risk models are evaluated. Model A has high training and validation error. Model B has very low training error but high validation error. Model C has low and similar training and validation errors.

a) Identify which model has high bias and which has high variance.

b) Explain how model complexity affects bias and variance.

c) Suggest a suitable improvement for Models A and B.

Evidence Expected

Correct identification, bias–variance explanation, comparison of model behaviour and suitable strategies.

Ans:

1. (a) Identifying the models

Table 13: Diagnosis of the three models (error rates are illustrative values that fit the description)

Model	Training error	Validation error	Gap	Diagnosis
A	28%	30%	2 points	**High bias** (underfitting): wrong even on the data it was trained on
B	2%	26%	24 points	**High variance** (overfitting): memorizes the training patients
C	8%	9%	1 point	Low bias and low variance: generalizes well

The two errors tell different things: the **training error** shows how well the model can fit (bias), and the **gap** between validation and training error shows how much the fit depends on the particular training sample (variance).

2. (b) Effect of model complexity

For a model $\hat{f}$ trained on a random sample, the expected squared error at a point x splits into three parts:

$$E[\,(y - \hat{f}(x))^2\,] = \text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

- **Bias** is the error from wrong assumptions: how far the average fitted model is from the truth. A simple model (a straight line for a curved relationship) cannot represent the pattern, so its bias is high however much data is used.
- **Variance** is how much the fitted model changes from one training sample to another. A very flexible model follows the noise in each sample, so its predictions for a new patient swing widely.
- σ^2 is irreducible noise in the data (measurement error, unrecorded factors) that no model can remove.
- **Increasing complexity lowers bias and raises variance.** Total error falls at first (bias dominates) and rises later (variance dominates), so the best model lies in between.

Demonstration. I fitted polynomials of degree 1 to 10 to 12 noisy observations of a smooth curve (noise standard deviation 0.25, so $\sigma^2 = 0.0625$), repeating the experiment on 400 different random samples and computing bias² and variance directly (Figure 4 and the table below).

Table 14: Bias² and variance by model complexity (400 simulated training sets)

Degree	Bias²	Variance	Noise σ²	Total expected error
1	0.2156	0.0103	0.0625	0.2885
3	**0.0061**	**0.0180**	**0.0625**	**0.0866**
5	0.0001	0.0270	0.0625	0.0896
8	0.0002	0.0511	0.0625	0.1137
10	0.0009	0.2313	0.0625	0.2947

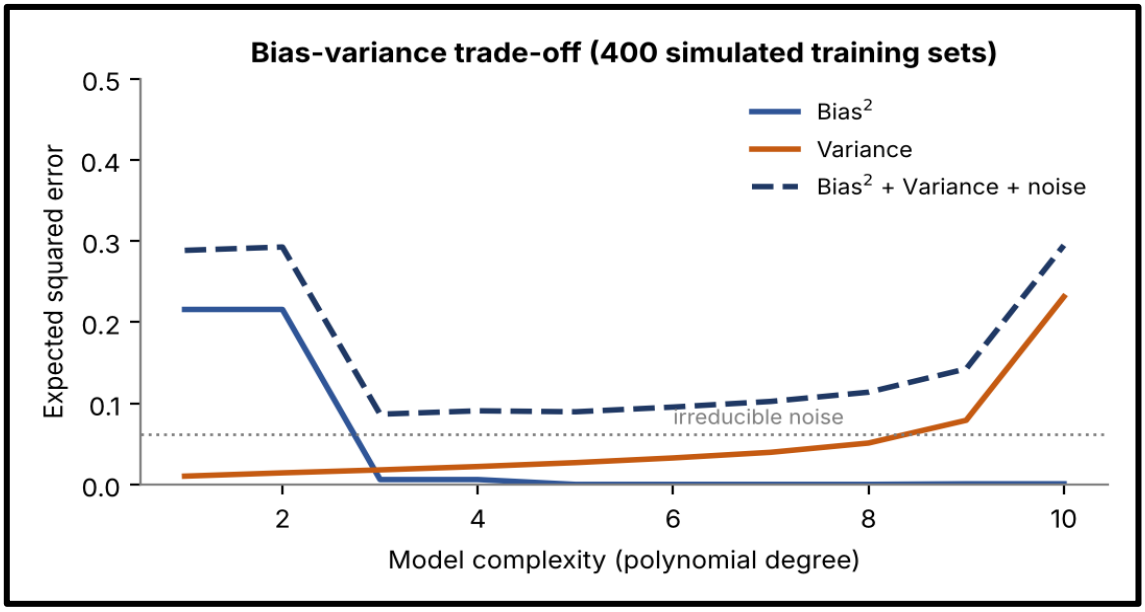


Figure 4: Bias falls and variance rises with complexity; total error is smallest in between

Degree 1 has bias² = 0.2156 and almost no variance (like Model A); degree 10 has almost no bias but variance 0.2313 (like Model B); degree 3 gives the smallest total (0.0866, the bold row), the balance reached by Model C.

3. (c) Improvements

Table 15: Remedies for Models A and B

Model	Problem	Suitable improvements	Will more data help?
A	High bias	Use a more flexible model (non-linear, deeper network, tree ensemble); add informative features or interaction terms (age × glucose); reduce regularization; train longer; check that features are normalized and the optimizer has converged	No: the model is too simple to use it
B	High variance	Collect more labelled patients; add regularization (L2/L1 penalty, dropout); simplify the model or stop training early; remove noisy features; use bagging or ensembles; choose complexity by cross-validation	Yes: the gap narrows as n grows

Learning curves (training and validation error against training-set size) show which case applies: for Model A the two curves converge at a high error; for Model B they stay far apart.

TASK 8 – Overfitting, Underfitting and Model Capacity 4 Marks

A simple model fails to capture disease patterns, a moderately complex model performs consistently, and a highly complex model performs almost perfectly on training data but poorly on unseen patients.

a) Identify underfitting, appropriate generalization and overfitting.

b) Explain how model capacity affects the ability to learn complex relationships.

c) Explain why increasing capacity may first improve performance but later cause overfitting.

Evidence Expected

Correct classification of model behaviour, explanation of capacity and relationship with generalization.

Ans:

1. (a) Classifying the three behaviours

Table 16: Model behaviour and its name

Model in the question	Behaviour	Term	Typical training / unseen error
Simple model	Fails to capture disease patterns	**Underfitting** (high bias)	High / high
Moderately complex model	Performs consistently on seen and unseen patients	**Appropriate generalization**	Low / similar
Highly complex model	Almost perfect on training data, poor on unseen patients	**Overfitting** (high variance)	Near 0 / high

2. (b) What model capacity is

Capacity is the range of functions a model can represent (its hypothesis space). It grows with the number of parameters and features, the polynomial degree, the depth and width of a network and the number of tree splits; regularization effectively lowers it. A **low-capacity** model (a line, i.e. logistic regression on raw features) can only learn a simple, near-linear boundary, so it misses a relationship such as "risk rises sharply when age is high **and** glucose is high". A **high-capacity** model can represent such interactions and almost any shape, so it can learn complex disease patterns, but it can equally represent the random noise in the training sample.

Demonstration. I generated 12 noisy training points from a smooth curve and 500 independent validation points (noise variance 0.0625) and fitted polynomials of increasing degree (Figure 5 and the table below).

Table 17: Training and validation error against capacity (mean squared error)

Model	Degree (capacity)	Training MSE	Validation MSE	Behaviour
Simple	1	0.293	0.300	Underfitting: both errors high
Moderate	**3**	**0.149**	**0.103**	**Good generalization: lowest validation error**
Very complex	11	≈ 0.000	3.55	Overfitting: 12 parameters interpolate 12 points, unseen error explodes

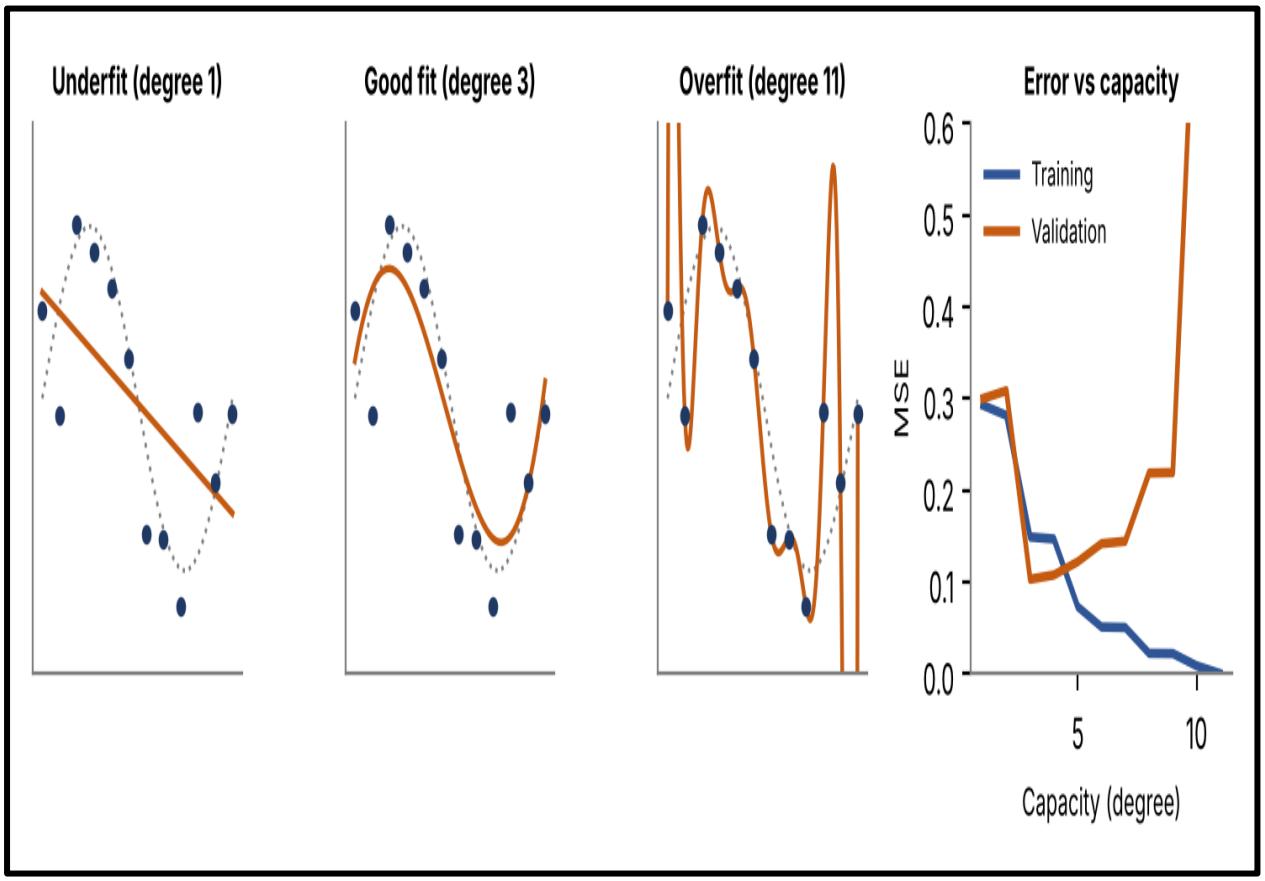


Figure 5: Fits of capacity 1, 3 and 11 (dotted = true curve) and training/validation error against capacity

The degree-3 model has a validation error of 0.103, close to the noise floor of 0.0625. (With only 12 training points its training error can exceed its validation error by chance; the general pattern is unaffected.)

3. (c) Why more capacity first helps and then hurts

1. **Stage 1: bias dominates.** While capacity is below the complexity of the true relationship, every added parameter lets the model represent more of the real pattern. Bias falls quickly while variance rises only a little, so **both training and validation error fall** (degree 1 → 3 above).
2. **Stage 2: the right capacity.** When the model can represent the true pattern, little bias is left to remove and the validation error reaches its minimum.
3. **Stage 3: variance dominates.** Extra capacity can only be used to fit the noise of the particular training sample. The training error keeps falling, reaching 0 when the parameters equal the number of points, but the model's predictions between and beyond the points oscillate, so the **validation error rises** (degree 5 → 11). The generalization gap (validation minus training error) widens.

This produces the U-shaped validation curve of Figure 5: the best model is at the **bottom of the validation curve, not the bottom of the training curve**. Capacity should therefore be chosen on validation data, and an over-large model can be tamed with more data, regularization, early stopping, dropout or cross-validation.

TASK 9 – Supervised Learning and Model Evaluation 3 Marks

A hospital has historical patient records labelled as diseased or disease-free and wants to predict outcomes for new patients.

a) Explain why this is a supervised learning problem.

b) Describe the roles of training and testing sets.

c) If 92 of 100 test patients are classified correctly, calculate accuracy and interpret it.

Evidence Expected

Identification of supervised learning, training/testing explanation, accuracy calculation and interpretation.

Ans:

1. (a) Why this is supervised learning

Every historical record is a pair (**x**, y): the patient's features **x** (age, blood pressure, glucose, cholesterol, family history) and a known **label** y ∈ {diseased, disease-free}. The task is to learn a function f: **x** → y that reproduces these labels and to apply it to new patients whose label is not yet known. The labels "supervise" learning through a loss function that measures the disagreement between prediction and label; because the label is discrete the problem is **classification** (a numeric output such as a risk score would be regression).

2. (b) Roles of the training and testing sets

Table 18: Data splits

Set	Typical share	Role	Rule
Training	60–80%	The model's parameters (for example the weights w) are fitted to minimize the loss on these labelled records	Normalization statistics are computed here only
Validation	10–20%	Used to choose hyper-parameters and the capacity (Task 8) and to detect overfitting	May be reused while tuning
Testing	10–20%	Held back and used once at the end to estimate how the final model performs on **unseen** patients	Never used for training or tuning

A model can score highly on records it has memorized (Task 8), so training accuracy says little about new patients. The test set imitates future patients and gives an honest estimate of **generalization**. The split should be random but stratified (same share of diseased patients in each part) and made **by patient**, and the test set must stay untouched; otherwise the estimate is optimistic (data leakage).

4. (c) Accuracy

$$\text{Accuracy} = \text{correct} / \text{total} = (TP + TN)/N = 92/100 = 0.92 = 92\%$$

The error rate is 1 − 0.92 = 8%. **Interpretation:** the model classifies 92 of every 100 unseen patients correctly. Three cautions go with this number:

- **Sampling uncertainty.** With only 100 test patients the standard error is $\sqrt{(0.92 \times 0.08/100)} = 0.027$, so an approximate 95% interval is 92% ± 5.3 points (about 87% to 97%).
- **Class balance.** If only 10% of the test patients are diseased, a useless model that labels everybody "disease-free" already scores 90%, so 92% is then only a small improvement.
- **Type of error.** A missed case (false negative) is usually costlier than a false alarm, so the confusion matrix must also be examined; one matrix that gives exactly 92% follows.

For example, TP = 40, FN = 5, FP = 3 and TN = 52 (45 diseased and 55 disease-free patients) give accuracy = 0.92, recall (sensitivity) = 40/45 = 0.889; precision = 40/43 = 0.930; specificity = 52/55 = 0.945. The model finds 88.9% of the diseased patients; the other 11.1% are missed, which accuracy alone hides.

TASK 10 – Unsupervised Learning and Patient Grouping 4 Marks

A hospital has patient records containing age, blood pressure, glucose and cholesterol but no predefined risk labels.

a) Explain why unsupervised learning is suitable.

b) Describe how K-means can group patients with similar characteristics.

c) Explain how distance can be used to measure similarity.

d) Describe how the clusters could support further clinical investigation without replacing medical diagnosis.

Evidence Expected

Learning approach, clustering procedure, distance-based similarity and meaningful interpretation.

Ans:

1. (a) Why unsupervised learning is suitable

2. (b) How K-means groups patients

K-means divides n patients into k clusters by minimizing the **within-cluster sum of squares**, $J = \Sigma$ over clusters c of Σ over patients x in c of $\|x - \mu_c\|^2$, where μ_c is the centroid (mean patient) of cluster c. The algorithm is:

1. Normalize the features (Task 1) and choose k (for example by the elbow method or the silhouette score).
2. Initialize k centroids (k-means++ or k random patients).
3. **Assignment step:** put every patient in the cluster of the nearest centroid (smallest Euclidean distance).
4. **Update step:** replace each centroid by the mean of the patients assigned to it.
5. Repeat steps 3 and 4 until the assignments no longer change. J never increases, so the procedure converges (to a local minimum, so several restarts are used).

Worked example. Six normalized patients (glucose, blood pressure, cholesterol), k = 2, starting centroids P1 and P5.

Table 19: Six normalized patients

Patient	Glucose	Blood pressure	Cholesterol
P1	0.20	0.25	0.30
P2	0.25	0.30	0.20
P3	0.15	0.20	0.25
P4	0.80	0.85	0.75
P5	0.85	0.80	0.90
P6	0.75	0.90	0.80

Table 20: Iteration 1: distances to the two centroids (c1 = P1, c2 = P5)

Patient	Distance to c1	Distance to c2	Assigned cluster
P1	0.000	1.042	1
P2	0.122	1.049	1
P3	0.087	1.128	1
P4	0.960	0.166	2
P5	1.042	0.000	2
P6	0.987	0.173	2

Update. Cluster 1 = {P1, P2, P3}, so c1 = ((0.20 + 0.25 + 0.15)/3, (0.25 + 0.30 + 0.20)/3, (0.30 + 0.20 + 0.25)/3) = (0.20, 0.25, 0.25). Cluster 2 = {P4, P5, P6}, so c2 = (0.80, 0.85, 0.817). **Iteration 2** uses the new centroids and assigns every patient to the same cluster as before, so the algorithm stops after 2 iterations with J = 0.0367.

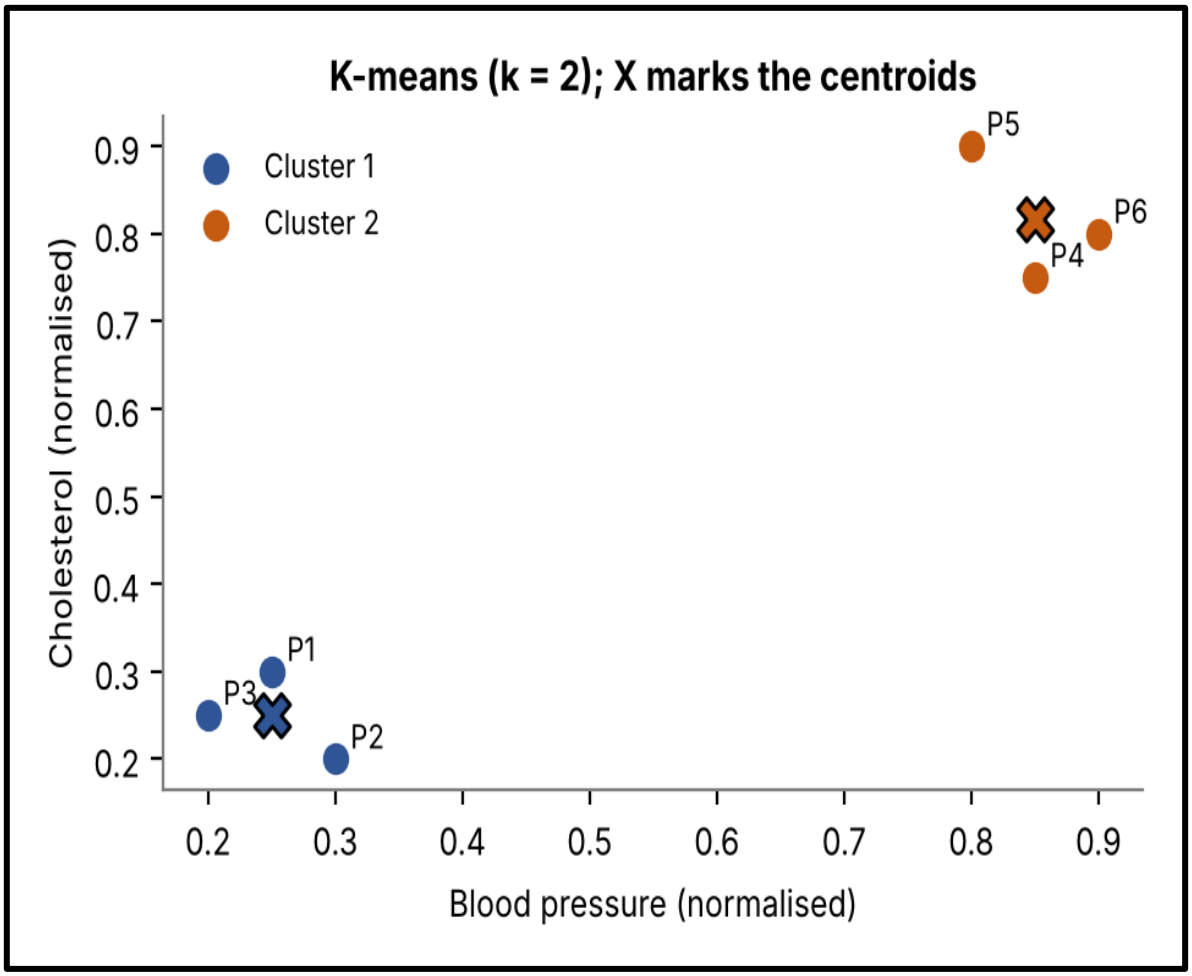


Figure 6: Result of K-means with k = 2 (two of the three features are plotted)

3. (c) Distance as the measure of similarity

K-means treats **small Euclidean distance as high similarity**. The distance $d(x, \mu) = \sqrt{\Sigma_j(x_j - \mu_j)^2}$ decides the assignment: for P4 the distances to the final centroids are 0.985 (c1) and 0.067 (c2), so P4 joins cluster 2. The centroid is the "average patient" of a group and the distance to it shows how typical a patient is; a patient far from every centroid is an outlier worth reviewing. Because distance adds the feature differences, the features must first be normalized (Tasks 1 and 6). Other measures (Manhattan distance, or Mahalanobis distance, which allows for correlated features such as blood pressure and age) can be used when they suit the data better.

4. (d) Supporting clinical investigation without replacing diagnosis

- **Describe each cluster by its profile** (mean glucose, blood pressure, cholesterol, age). Here cluster 2 has high values on all three features (about 0.80, 0.85 and 0.82) and cluster 1 low values (0.20, 0.25 and 0.25). Clinicians can name the first a "high metabolic load" group and invite its members earlier for screening, lifestyle advice or confirmatory tests, using the groups to **prioritize** attention.
- **Generate hypotheses and labels.** Check whether disease rates differ between clusters in the labelled subset, and sample a few patients from each cluster for clinical review, which is a cheap way to create labels for Task 9.
- **Keep the doctor in charge.** A cluster is a statistical grouping, not a disease: it was found without any knowledge of diagnosis, the groups depend on k, the initialization and the features, and a patient near a boundary may be misplaced. Clusters should be validated on new data (stability), checked for fairness across age and sex groups, and used only to **support** a diagnosis made from clinical examination and tests.

TASK 11 – Comparison of Learning Approaches 4 Marks

Dataset A contains patient features with disease labels, while Dataset B contains patient features without labels.

a) Identify the suitable learning paradigm for each dataset and justify your choice.

b) Compare their objectives and outputs.

c) Recommend an approach for predicting disease status in a new patient and justify the recommendation.

Evidence Expected

Correct identification, comparison of objectives and outputs, and context-based justification.

Ans:

1. (a) Paradigm for each dataset

- **Dataset A (features + disease labels): supervised learning.** The labels give the correct answer for every example, so the model can be trained to map features to the label and be tested against known answers (Task 9).
- **Dataset B (features only): unsupervised learning.** With no labels there is no answer to imitate, so the only thing that can be learnt is the structure of the features: clusters, outliers or a compact representation (Task 10).

2. (b) Comparison of objectives and outputs

Table 21: Supervised versus unsupervised learning

Aspect	Supervised (Dataset A)	Unsupervised (Dataset B)
Objective	Learn f(x) ≈ y; minimize the prediction error against known labels	Discover structure in x: groups, density, outliers, low-dimensional structure
Data needed	Features **and** labels (costly to obtain)	Features only (plentiful)
Output	A predicted class or risk probability P(disease | x) for a new patient	Cluster membership, centroids, distances, a reduced representation
Typical methods	Logistic regression, decision trees, neural networks	K-means, hierarchical clustering, PCA, autoencoders
Evaluation	Accuracy, recall, precision, AUC on a test set with known labels	No ground truth: silhouette, within-cluster sum of squares, stability, clinical review
Answers the question	"Will this patient have the disease?"	"Which patients are alike and are there unknown subgroups?"
Main risk	Overfitting; labels that are few, noisy or biased	Groups that are mathematically neat but clinically meaningless

3. (c) Recommended approach for a new patient

Recommendation: a supervised classifier trained on Dataset A, starting with regularized logistic regression on standardized features as a transparent baseline and moving to a higher-capacity model (decision-tree ensemble or small neural network) only if the validation error shows that the baseline underfits. The output is P(disease | x), which can be used with a threshold or ranked (Tasks 3 and 4).

- **It optimizes the quantity actually wanted.** The target is disease status, and only labels can tell the model what disease looks like. A clustering of Dataset B can say that a patient resembles group 2 but not that group 2 is diseased.
- **Its quality can be measured** on a held-out test set (Task 9), and it **gives probabilities** that combine with test results by Bayes' rule (Task 4); an unsupervised method has neither.

Dataset B should still be used in a supporting role: to fit normalization statistics on more patients, to discover subgroups to check separately, to flag unusual patients and, if labels are scarce, for semi-supervised learning. This combination is the basis of Task 12.

TASK 12 – Design Challenge: Adapting the Strategy 8 Marks

Changed Requirement

The hospital initially analyzes **10,000** patient records. It must now process **1,000,000** records, predict disease risk, discover patient groups, and handle missing values and differences in feature scales.

Decide whether to **retain, modify or redesign** the analytical strategy.

How I read the requirement. The **original strategy** is the one built in Tasks 1–11: standardize the features, train a supervised classifier on the labelled records (Task 11), and run K-means separately on 10,000 complete records (Task 10), all in memory with exact calculations. The change makes the data **100 times larger** and adds three demands: group discovery together with prediction, missing values, and different feature scales. I answer with **MODIFY**: keep the mathematical core (standardization, a probabilistic classifier, K-means) but change how the parts are connected and computed.

A. Impact Analysis 2 Marks

a) Explain how increasing the dataset size affects data representation, numerical computation and learning efficiency.

b) Discuss why normalization, missing-value handling and data variability are important.

Evidence Expected

A clear analysis of how the data scale and data quality affect the original design.

Ans:

I measured the growth directly. A simulated cohort of 1,000,000 patients with the four features was generated in the program (Appendix A) and one gradient step of logistic regression and one K-means iteration (k = 3) were timed on the first 10,000 and on all 1,000,000 records (one run on a single workspace core; the figures show the growth, not production speed).

Table 22: Original size compared with the new requirement

Quantity	10,000 records	1,000,000 records	Factor
Feature matrix (4 features, 8 bytes each)	320 KB	32 MB	100×
One gradient step (cost O(nd))	0.23 ms	21.1 ms	91× (measured)
One K-means iteration, k = 3 (cost O(nkd))	0.44 ms	60.4 ms	138× (measured)
Matrix of all pairwise distances ($n^2 \times 8$ bytes)	800 MB	8 TB	10,000×
Patients left if each feature is missing 5% of the time and incomplete records are dropped	8,145 of 10,000	814,506 of 1,000,000	81.5% kept

a) Representation, numerical computation and learning efficiency

- **Representation.** The feature matrix grows from 40,000 to 4,000,000 numbers (32 MB), which still fits in memory; structures that grow with n^2 do not. The 10^{12} pairwise distances of exact k-nearest-neighbour search or hierarchical clustering would need 8 TB and are no longer an option, so algorithms must be linear in n. Family history is a yes/no feature and is stored as 0/1.
- **Numerical computation.** Adding a million floating-point numbers accumulates rounding error, so means and variances must be computed with float64 and a stable method (pairwise or Welford summation). One full gradient step now touches every record (about 91× more time in the measurement), so training changes to **mini-batch** gradient descent, whose cost per step does not depend on n, and K-means changes to a mini-batch version.
- **Learning efficiency.** More data helps the statistics: the standard error of an estimated accuracy falls by $\sqrt{100} = 10$ (Task 9), and the variance part of the error falls (Task 7), so a higher-capacity model can now be afforded (Task 8). The price is computing time, which is why the cost of each step matters.

b) Normalization, missing values and variability

- **Normalization.** The simulated features have standard deviations 14.0, 18.0, 31.2, 38.0 (age, blood pressure, glucose, cholesterol). In raw K-means the share of the total variance, which drives the distances, would be 6.8%, 11.3%, 32.1%, 49.8%, so **cholesterol and glucose alone would decide 82% of every distance** (Tasks 1 and 6). Standardizing reduces the condition number of the feature covariance from 7.38 to 1.005, which speeds up gradient descent. The means and standard deviations must be fitted once on training data and reused unchanged in deployment.
- **Missing values.** If each of four features is missing independently in 5% of records, only $0.95^4 = 81.5\%$ of records are complete: deleting the others throws away **about 185,000 patients**. Worse, values are rarely missing at random (glucose may be measured only when a patient looks unwell), so deletion biases the data. Missing values should be imputed (median or model-based), with an indicator "was missing", and the imputer fitted on training data only.
- **Variability.** Records come from different clinics, devices and populations; glucose is right-skewed with outliers (the simulated values are log-normal). Such differences inflate the spread, distort the mean used in standardization and shift the distribution between training and later use. Robust scaling (median and inter-quartile range), a log transform for glucose, clipping of impossible values and monitoring for distribution shift keep the models stable.

B. Evaluation of the Original Approach — 2 Marks

a) Evaluate whether a single supervised approach is sufficient for both disease prediction and patient group discovery.

b) Explain the limitations of labelled learning when useful patterns also exist in unlabelled records.

Evidence Expected

Evaluation of the original strategy against the changed needs, with its limitations.

Ans:

a) A single supervised approach is not sufficient

Table 23: What each need requires

Need	Supervised model on labelled records	Needs unsupervised learning?
Predict disease risk	**Yes**: this is exactly what labels support	No
Discover patient groups	**No**: the classifier is trained to separate two classes, and a decision boundary says nothing about subgroups inside a class or groups that no label describes	**Yes**: K-means or other clustering of the features
Use the unlabelled records	**No**: they have no y and are ignored by the loss function	**Yes**: they shape the clusters and the scaling
Explain who is at risk	Partly: a probability, not a description of the patient type	Cluster profiles describe patient types

A supervised model answers "what is P(y | x)?". Group discovery asks "what is the structure of x?". These are different questions, so one supervised model cannot supply both; the original design had the right ingredients but treated them as separate tasks on a small file.

b) Limits of labelled learning when unlabelled records contain patterns

- **Few labels.** A label needs a confirmed diagnosis; in a hospital file the labelled records may be a small fraction of the 1,000,000. A high-capacity model trained on few labels overfits (Task 8), yet most of the data are never used.
- **Biased labelled sample.** Patients who were tested are often those already suspected to be ill, so the labelled group is not representative. The unlabelled records show what the wider population looks like and expose the shift.
- **Hidden structure is lost.** If a group of patients (say high blood pressure with low glucose) is common in the unlabelled data but absent from the labelled sample, the classifier has no basis for rating it, and a new subtype can never be found by a model that is only shown known labels.
- **Information about p(x) is thrown away.** Semi-supervised learning rests on the cluster assumption: patients in the same dense cluster tend to share a label, so unlabelled points show where the decision boundary should not cut.

C. Adaptation or Redesign — 2 Marks

a) Propose a strategy combining supervised classification and unsupervised clustering.

b) Explain how probability, distance measures, normalization and model capacity can be incorporated.

Evidence Expected

A justified combined strategy and the way the mathematical tools are built into it.

Ans:

a) Proposed strategy: one pipeline with a supervised and an unsupervised branch

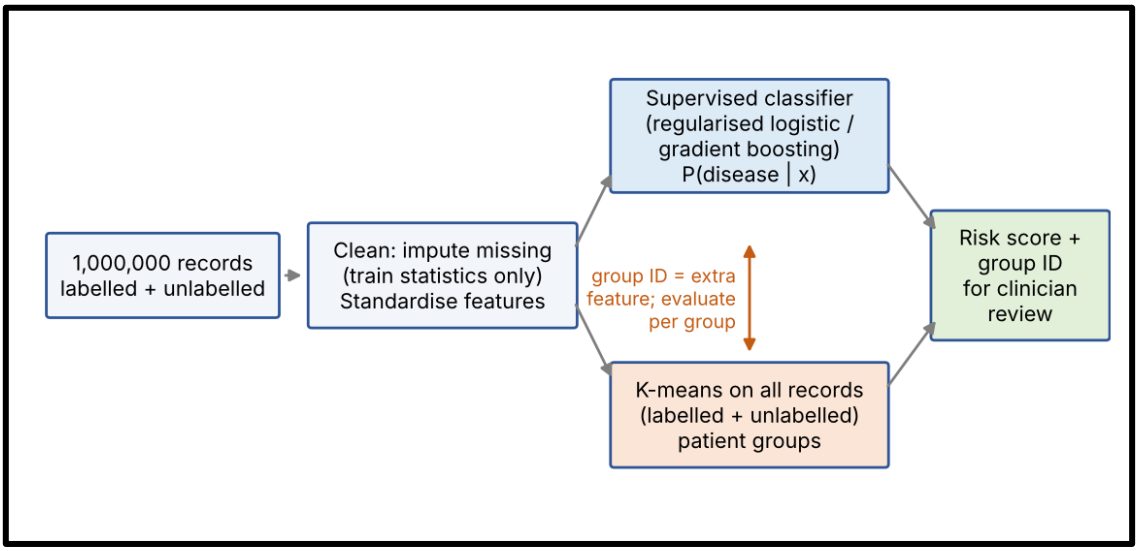


Figure 7: Proposed design: shared cleaning, then classifier and clustering feeding one clinician report

1. **Clean and standardize (shared).** Impute missing values and add "missing" flags; apply robust scaling or a log transform where needed; standardize with training statistics.
2. **Cluster all records.** Run mini-batch K-means (k chosen by silhouette and clinical sense) on labelled **and** unlabelled records, giving each patient a group ID and a distance to the nearest centroid.
3. **Train the classifier on labelled records** with the original features plus the group ID, using mini-batch gradient descent and validation-chosen capacity. Optionally pseudo-label confident unlabelled patients inside pure clusters (semi-supervised) and retrain.
4. **Report** the risk probability, the group and the distance to the centroid (out-of-distribution warning) to the clinician, who decides.

The design is a **modification**, not a replacement: every component is a method already analysed in Tasks 1–11.

b) Incorporating the mathematical tools

Table 24: Where each tool enters the design

Tool	Where	How
Probability	Classifier output; combining evidence	The logistic model gives $P(D \mid x) = \sigma(w^T x + b)$; a prevalence change at another hospital is handled by re-weighting the prior; a lab test is added as a likelihood ratio (Task 4); the decision threshold comes from the costs of missed cases
Distance	Clustering; outlier warning	Euclidean distance on standardized features assigns patients to centroids; a large distance to every centroid flags an unusual patient; the same distance supports a k-nearest-neighbour check on a sample

Tool	Where	How
Normalization	First step of both branches	Z-score (robust version for glucose) with training statistics only; avoids dominance of cholesterol and glucose and improves the conditioning of gradient descent
Model capacity	Classifier and k	Increase capacity from logistic regression to a tree ensemble or small network while the validation loss falls (Task 8); use L2 penalty and early stopping; choose k by silhouette because k = n gives zero error but no insight; more data ($n = 10^6$) justifies higher capacity
Information gain	Feature selection	Rank features by information gain (Task 5) to drop uninformative inputs and reduce variance

D. Evidence-Based Defence — 2 Marks

a) Propose suitable experiments and evaluation measures to compare the redesigned approach with the original.

b) Justify the final design using mathematical reasoning, generalization, computational needs and data characteristics.

Evidence Expected

An evaluation plan and a justification supported by mathematics, generalization, computation and data.

Ans:

a) Experiments and evaluation measures

Table 25: Proposed evaluation plan (original design = Tasks 1–11 pipeline on complete records; redesign = Part C)

Experiment	What is compared	Measures
E1 Held-out comparison	Original and redesign on the same patient-level, stratified train/validation/test split, repeated with 5-fold cross-validation	AUC, recall at a fixed specificity, precision, F1, log-loss and Brier score; calibration curve; 95% bootstrap interval of the difference
E2 Ablation	Remove one change at a time: imputation (use complete cases), standardization, cluster features, mini-batch training	Change in AUC and recall; shows which change is responsible for the gain
E3 Capacity and data size	Models of rising capacity; training sizes 10^4, 10^5, 10^6	Training and validation curves; point where the gap stops closing (Tasks 7 and 8)
E4 Cluster quality	K-means with and without scaling; k = 2 to 10	Silhouette, Davies–Bouldin index, stability across seeds (adjusted Rand index), disease rate per cluster, clinician review

Experiment	What is compared	Measures
E5 Robustness and subgroups	Performance per cluster, hospital, age band; with 10% extra values removed	Per-group recall and calibration; drop in AUC under missingness
E6 Computational cost	Run time and memory against n = 10^4, 10^5, 10^6	Seconds per epoch / per K-means iteration, peak memory

Decision rule. Adopt the redesign if the 95% interval of its AUC (or recall at fixed specificity) improvement over the original is above zero, calibration is not worse, no subgroup is worse off, and the training time fits the hospital's overnight budget. Otherwise keep the original and modify only the components that passed E2.

b) Justification of the final design

- **Mathematical reasoning.** Standardization lowers the condition number from 7.38 to 1.005 in the 1,000,000-patient simulation, so gradient descent needs fewer steps; K-means needs it so that its Euclidean distances are not decided by two features (Task 6). Posterior probabilities from a calibrated model (Bayes, Task 4) allow the threshold to follow the cost of errors and the local prevalence.
- **Generalization.** With 100 times more data the variance term of the error is smaller (Task 7), so higher capacity is justified, but it is chosen on validation data and controlled by regularization (Task 8). The held-out test set and per-group checks measure whether this really holds for unseen patients.
- **Computational needs.** Every step is linear in n. At 1,000,000 records one K-means iteration took 60.4 ms, so 50 iterations take about 3.0 s, and mini-batch gradient descent has a cost per step that does not depend on n. Exact all-pairs methods would need about 10^{12} distances and 8 TB and are therefore excluded.
- **Data characteristics.** Most records are unlabelled, so clustering and semi-supervised use of them is the only way to benefit from the whole file; imputation keeps the 185,000 patients that complete-case analysis would discard and avoids the selection bias of dropping them; robust scaling copes with skewed, outlying glucose values.

Table 26: Final decision: retain, modify or redesign each component

Component	Decision	Reason
Standardization of features	**Retain** (fit on training data only)	Needed for distances and conditioning; add a robust variant for glucose
Supervised probabilistic classifier	**Retain and modify**	Add mini-batch training, validation-chosen capacity, regularization and the cluster features
K-means	**Retain and modify**	Use mini-batch version on all records; choose k by silhouette
Complete-case analysis	**Redesign**	Replace by imputation with missing flags (keeps the 18.5% of patients that would be dropped and reduces bias)
Exact pairwise calculations	**Redesign**	n^2 memory and time are impossible at $n = 10^6$
Evaluation	**Modify**	Add calibration and per-group results

Conclusion of Task 12: MODIFY. The foundations survive the hundred-fold increase in size; what changes is the data preparation, the computation (mini-batch, linear cost) and the combination of supervised and unsupervised learning.

Overall Conclusion

The assignment asked how mathematical foundations decide whether a learning system for healthcare risk can be trusted. The calculations lead to five conclusions. First, **scale decides what a model sees**: on raw data glucose and cholesterol supplied 92% of the squared distance between two patients, and standardization made the four features contribute comparably (Tasks 1, 2 and 6). Second, **probability keeps evidence honest**: a test that finds 90% of cases still gives only a 66.7% chance of disease after a positive result when prevalence is 10%, and elevated glucose raises the risk from 20% to 57.1% (Tasks 3 and 4). Third, **entropy quantifies uncertainty** (0.971 bits for the 40/60 risk split) and information gain ranks the features that remove it (Task 5). Fourth, **generalization is a balance**: bias falls and variance rises with capacity, and the best model sits at the bottom of the validation curve (Tasks 7 and 8). Fifth, **labelled and unlabelled data call for different tools**, a supervised classifier to predict and K-means to discover groups (Tasks 9–11), and for 1,000,000 records the best design **modifies** the original by combining both, imputing missing values and using linear-cost mini-batch computation (Task 12).